

Appendices

Appendix A: Initial Project Proposal

Initial Project Specification (BSc 2025-26)

Student Number: 2604383

Proposed Title: Forecasting UK Frequency Grid Instability: A Data-Driven Alert System

Proposed Aim: Predict UK grid instability from weather and frequency data for proactive renewable energy alert system.

Proposed Objectives:

By the end of this project, I will be able:

1. To research and select a robust proxy variable (e.g., System Frequency deviation $> \pm 0.2$ Hz) for grid instability using National Grid ESO data.
2. To build a demo of data pipeline to integrate, clean, and feature-engineer National Grid ESO and HadUK-Grid datasets using polars.
3. To benchmark three forecasting models (SARIMAX, LightGBM, LSTM) using RMSE, MAE, and training time.
4. To incorporate probabilistic forecasting (quantile regression) to support uncertainty-aware alerts.
5. To develop a proof-of-concept decision-support system integrating a rule-based alert engine and Streamlit dashboard.
6. To validate the alert system on historical data, including the August 9, 2019 blackout event.
7. Critically evaluate the system's performance using SHAP values and assess deployment viability.

Proposed Rationale:

The UK's transition to renewable energy introduces significant grid volatility. Currently, grid operators use costly, high-carbon reactive measures to prevent blackouts. This project addresses this gap by developing a proactive solution that forecasts instability before it happens, allowing for planned, cost-effective, and low-carbon interventions.

Facilities Required:

Python (Polars, Pandas, Scikit-learn, LightGBM, TensorFlow), Streamlit, HadUK-Grid, National Grid ESO API.

Appendix B: Final Project Proposal & Scope Evolution

Final Project Specification (GridGuardian)

Final Title: GridGuardian: Proactive Grid Instability Forecasting using Physics-Informed Machine Learning

Final Aim: Develop a robust, real-time alert system using a hybrid ensemble (LightGBM + LSTM) and physics-informed features to provide proactive foresight for UK grid frequency stability.

Evolution of Methodology:
The project evolved from the initial proposal’s focus on SARIMAX to a Hybrid LGBM-LSTM Ensemble. SARIMAX was found to be insufficient for capturing the non-linear, high-frequency transients observed during grid disturbances (e.g., the August 2019 blackout). The final implementation incorporates physics-informed features (Swing Equation logic, OpSDA wind compression) and a multi-signal alert fusion engine to provide a more robust “Command Deck” solution.

Achieved Technical Deliverables:

Original Objective	Final Implementation Outcome
1. Select robust proxy	Established instability threshold at 49.85 Hz with physics-informed triggers (RoCoF & Acceleration).
2. Data pipeline	Engineered a high-performance Polars ETL pipeline merging 1s frequency, HadUK weather, and inertia data.
3. Benchmark models	SARIMAX, LightGBM, and LSTM benchmarked; Result: Hybrid LGBM-LSTM selected for superior precision.
4. Probabilistic alerts	Implemented Quantile Regression ($\alpha=0.1$ to 0.9) with post-hoc Isotonic Calibration for reliable uncertainty.
5. Streamlit Dashboard	Developed “Command Deck” v3 with split-cache design and SHAP-based real-time risk driver analysis.
6. Blackout Validation	Validated against August 9, 2019 event; demonstrated proactive lead-time warning of >10 seconds.
7. Critical Evaluation	Conducted SHAP global/local importance analysis and out-of-season robustness testing on Winter data.

Final System Rationale:
By moving beyond simple forecasting to a physics-aware “Command Deck,” the GridGuardian system demonstrates that data-driven foresight can significantly reduce reliance on carbon-intensive reactive measures, directly supporting the UK’s transition to a high-renewables grid.

Appendix C: Ethical Approval

This project utilizes publicly available open datasets from National Grid ESO and Open-Meteo, classifying it as primary quantitative analysis. As the study does not involve human subjects or the collection of sensitive personal data, it qualifies for the internal ethical approval process for non-human research.

The following documentation confirms the internal ethical clearance obtained for the study:

11/26/25, 11:55 AM

CN6000 Internal Ethical Approval Process 2025

ETHICS CN6000 Internal Ethical Approval Process 2025

Only complete this form if you have already received approval for the Proposal. If you complete this form before getting the final approval of your proposal, this form will be deleted, and you will be required to do it again!

Do you need internal approval?

This section aims to check if you need internal approval in the first place. If you are not applying field research, then you may not need internal or external approval.

1. Have you received the final Proposal Approval? That is the proposal has been reviewed by the 'second supervisor' and it has been approved? *

☒ Yes
☐ No
☐ Maybe

2. My study will not involve collecting any data from humans. It is either just based on published datasets that I did not create or there are no datasets at all part of my study. *

☒ True
☐ False

<https://forms.office.com/Pages/ResponseDetailPage.aspx?d=aaKUX5eud0eRi0Z2BGW1cS2Lw3259NmgdTAKKknKJUMU8yVThTWVRaV0ZQWV>

Figure 1: Internal screening confirming the use of public datasets and absence of human data collection.

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3. Please state your topic title: * 

Illuminating the black box: an explainable AI framework for brain tumor segmentation using volumetric data interpretation in 3D CNNs.

4. Please state your supervisor's name (Spell it correctly). * 

Mimoona Maqsood SHARIF

5. If you answered **Yes** to question 1 and **True** to Question 2 - You are not required to proceed. Just click NEXT then Submit without the need to answer the remaining questions. Otherwise, please proceed to section 2. * 

☒ True

☐ False

This section aims to see if we can approve your proposal internally. 

Please read the questions carefully. Feel free to discuss these questions with your supervisor. If you find out that you have to apply for approval by the Research Ethics Committee, please discuss with your supervisor if there is a way to modify your dissertation in a way that does not take away from the value of your work.

6. I intend to plan and conduct my study as a pilot test of the research only. 

☐ Yes

☐ No

<https://forms.office.com/Pages/ResponseDetailPage.aspx?id=aakUv5aud0eiRi0Z2BGW1qS2Lw3259NmgdTaKKknKJUMUbyVTnTWWRaV0ZQWV...>

Figure 2: Registration of the research topic and supervisor oversight for the data science project.

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☐ Maybe

7. I intend to conduct my field research as pilot test on UEL students or staff only. ☐

☐ Yes

☐ No

☐ Maybe

8. I will debrief participants before the study and mention that their contribution is entirely voluntary. ☐

☐ Yes

☐ No

☐ Maybe

9. I will not be collecting any personal data such as names, addresses, telephone numbers, emails...etc. ☐

☐ Yes

☐ No

☐ Maybe

10. I will not be asking any sensitive questions that can cause distress - such as mental health related or emotional questions. ☐

☐ Yes

<https://forms.office.com/Pages/ResponseDetailPage.aspx?id=aakUv5aud0eiRi0Z2BGW1qS2Lw3259NmgdTaKKknKJUMUbyVTnTWWBaV0ZQWV...>

Figure 3: Declaration of compliance with field research and participant debriefing standards.

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☐ No

☐ Maybe

11. I will be getting approval of the format and content of my questions from my supervisor before I proceed with my field research. ☐

☐ Yes

☐ No

☐ Maybe

12. I will have a 'consent' form for participants to be part of their study and be informed of their rights. ☐

☐ Yes

☐ No

☐ Maybe

13. I will store the data I collect on UEL OneDrive. No other copy of it will be stored elsewhere. ☐

☐ Yes

☐ No

☐ Maybe

14. I will not use the data for any other intended purpose and delete it at the end of the term. ☐

<https://forms.office.com/Pages/ResponseDetailPage.aspx?id=aakUv5aud0eiRi0Z2BGW1qS2Lw3259NmgdTaKKknKJUMUbyVTnTWWBaV0ZQWV...>

Figure 4: Confirmation of UEL data storage policies and ethical data usage protocols.

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☐ Yes

☐ No

☐ Maybe

15. If you have answered No or Maybe to any of the questions above, you cannot proceed with internal Ethical approval. I would advise you to review those particular questions with your supervisor before proceeding. If you both agree that you want to keep these answers, then please indicate below that you are happy to proceed. 

☐ All my answers are Yes. I have completed the internal approval and promise to abide by it.

☐ Not all my answers are yes. I have spoken to my supervisor and we both agreed that I should apply for form B Ethical Approval process.

16. I understand that violating any of the terms listed above could result in an automatic fail and possibly academic misconduct proceedings. 

☐ I confirm

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Figure 5: Final submission and confirmation of the internal ethical approval process.

Appendix D: Dashboard Outputs

Note: This section is reserved for high-resolution exports and annotated screenshots of the GridGuardian dashboard during various simulated and historical instability events (e.g., August 9, 2019 blackout, winter peak demand scenarios).

Appendix E: Code Snippets

This appendix presents the core implementation logic for the GridGuardian system, providing a technical reference for the feature engineering, probabilistic modeling, and real-time dashboard engine.

E.1: Physics-Informed Feature Engineering

The following snippets detail the physics-informed feature engineering pipeline, including the Optimised Swinging Door Algorithm (OpSDA) implementation and the derivation of causal multi-window RoCoF and wind ramp rates.

Optimised Swinging Door Algorithm (Implementation: opsda.py)

```
def compress(data, width):
    """
    Compresses a list of (timestamp, value) tuples using the Swinging Door Algorithm.
    """
    if not data:
        return []

    compressed_data = [data[0]]
    start_point_index = 0

    for i in range(1, len(data)):
        current_point = data[i]
        pivot_point = data[start_point_index]

        # Form a "door" from the pivot point to the current point
        upper_bound_slope = (pivot_point[1] + width - current_point[1]) /
(pivot_point[0] - current_point[0]) if pivot_point[0] != current_point[0] else
float('inf')
        lower_bound_slope = (pivot_point[1] - width - current_point[1]) /
(pivot_point[0] - current_point[0]) if pivot_point[0] != current_point[0] else
float('-inf')

        # Check all intermediate points
        for j in range(start_point_index + 1, i):
            intermediate_point = data[j]
            slope = (pivot_point[1] - intermediate_point[1]) / (pivot_point[0] -
intermediate_point[0]) if pivot_point[0] != intermediate_point[0] else float('inf')

            if slope > upper_bound_slope or slope < lower_bound_slope:
                compressed_data.append(data[i-1])
                start_point_index = i - 1
                break

    compressed_data.append(data[-1])
    return compressed_data
```

Wind Ramp Rate and Causal RoCoF (Selection from: feature_engineering.py)

```
def calculate_wind_ramp_rate(df):
    weather_data = df[["timestamp",
"wind_speed"]].drop_duplicates(subset=["timestamp"]).copy()
    weather_data['unix_ts'] = weather_data['timestamp'].astype(np.int64) //
1_000_000_000

    data_tuples = list(weather_data[['unix_ts',
```

```

'wind_speed']].itertuples(index=False, name=None))
    compressed = opsda.compress(data_tuples, width=OPSDA_WIDTH)

    compressed_df = pl.DataFrame(compressed, schema=["unix_ts", "wind_speed"],
orient="row")
    # Calculate ramp rate (slope between compressed points)
    compressed_df = compressed_df.with_columns(
        ((pl.col("wind_speed").diff()) /
(pl.col("timestamp").diff().dt.total_seconds()))).alias("wind_ramp_rate")
    )
    return df.merge_asof(compressed_df, on="timestamp", direction="backward")

def create_features(df):
    # Causal RoCoF and Acceleration
    df["rocof_1s"] = df["grid_frequency"].diff(1).fillna(0)
    df["rocof_5s"] = ((df["grid_frequency"] - df["grid_frequency"].shift(5)) /
5.0).fillna(0)
    df["rocof_accel"] = (df["rocof_5s"].diff(5) / 5.0).fillna(0)

    # Physics-informed risk signals
    df["wind_power_proxy"] = np.clip(df["wind_speed"]**3 * 3.0, 0, 3000)
    df["renewable_penetration_ratio"] = df["wind_power_proxy"] /
df["demand_proxy"].replace(0, 35000)
    df["rocof_inertia_risk"] = df["rocof_smooth"].abs() *
df["renewable_penetration_ratio"]
    return df

```

E.2: Probabilistic Model Logic

This section presents the probabilistic model logic, showcasing the quantile regression setup with weighted sampling to prioritize tail-risk events and the post-hoc isotonic recalibration routine.

Quantile Regression and Weighted Sampling (Selection from: `model_trainer.py`)

```

def train_quantile_model(df, alpha):
    params = LGBM_QUANTILE_PARAMS.copy()
    params['objective'] = 'quantile'
    params['alpha'] = alpha

    model = lgb.LGBMRegressor(**params)

    # Upweight samples where target frequency is low (approaching instability)
    sample_weights = np.where(y_train < 49.95, 10.0,
                             np.where(y_train < 50.00, 3.0,
                             1.0))
    model.fit(X_train, y_train, sample_weight=sample_weights)
    return model

```

Post-hoc Quantile Recalibration (`calibration.py`)

```

def fit_calibrator(y_true, y_pred_quantile, alpha):
    """Post-hoc recalibration using isotonic regression."""
    calibrator = IsotonicRegression(out_of_bounds='clip')
    calibrator.fit(y_pred_quantile, y_true)
    return calibrator

def calibrate_predictions(calibrator, y_pred_quantile):
    return calibrator.predict(y_pred_quantile)

```

E.3: Dashboard Engine & Multi-Signal Alert Logic

The dashboard engine logic illustrates the split-cache architecture for high-performance monitoring and the multi-physics alert fusion engine that powers the real-time Command Deck.

Cache Design and Alert Fusion (Selection from: `app.py`)

```
# Split-Cache Architecture
@st.cache_resource
def load_models(model_stamp):
    """Persists heavy model objects across user sessions."""
    lower_model = joblib.load(LGBM_QUANTILE_LOWER_PATH)
    lstm_model = tf.keras.models.load_model(LSTM_MODEL_PATH)
    return lower_model, lstm_model

# Variable Extraction from Current State
freq_now = current_row['grid_frequency']
rocof_now = current_row.get('rocof_smooth', 0.0)
rocof_accel = current_row.get('rocof_accel', 0.0)
volatility = current_row.get('volatility_10s', 0.0)
ren_pen = current_row.get('renewable_penetration_ratio', 0.0)

# Physics-Informed Multi-Signal Alert Fusion
rocof_alert = (rocof_now < -0.012) and (freq_now < 50.05)
accel_alert = (rocof_accel < -0.005)
volatility_alert = (volatility > 0.018) and (freq_now < 50.1)

# Score-based fusion (convergence of signals)
signal_count = sum([
    rocof_alert, accel_alert, volatility_alert,
    renewable_stress, freq_boundary,
    classifier_prob > 0.30, lstm_prob > 0.30, quantile_signal,
])

emergency_trigger = signal_count >= 3 or freq_now < alert_threshold_hz or
quantile_signal

# Intervention Simulator (Swing Equation Implementation)
#  $\Delta f = (\Delta P \times f_0) / (2 \times H \times S_{base})$ 
if synthetic_inertia_mw > 0:
    swing_delta_f = (synthetic_inertia_mw * NOMINAL_FREQ) / (2 * SYSTEM_INERTIA_H *
TOTAL_SYSTEM_CAPACITY)
    lower_bound_pred = lower_bound_raw + swing_delta_f
```

E.4: Global Configuration

The global configuration file defines the system thresholds, physics constraints, and model hyperparameters used throughout the GridGuardian pipeline.

System Thresholds and Feature Set (`config.py`)

```
TTA_SECONDS = 10
OPSDA_WIDTH = 0.5
LAG_INTERVALS_SECONDS = [1, 5, 60]

LGBM_FEATURE_COLS = [
    "grid_frequency", "rocof_smooth", "rocof_accel",
    "wind_speed", "wind_ramp_rate", "renewable_penetration_ratio",
    "inertia_value", "inertia_roc", "low_inertia_flag"
```

```
] + [f"lag_{lag}s" for lag in LAG_INTERVALS_SECONDS]
```

```
LGBM_QUANTILE_PARAMS = {  
    'n_estimators': 300,  
    'learning_rate': 0.03,  
    'max_depth': 6,  
    'reg_alpha': 0.1,  
    'reg_lambda': 1.0,  
}
```